

Evanns Morales-Cuadrado

U.S. Citizen

Robotics PhD Student Georgia Institute of Technology

CONTACT	Email: evannsmc@gmail.com Research Website: https://www.evannsmc.com/ LinkedIn: https://www.linkedin.com/in/evanns-morales/		Github: https://github.com/evannsmc
TECHNICAL SKILLS	• Programming Languages: Python, C/C++, Matlab/Simulink • Frameworks & Libraries: JAX, PyTorch, acados, CasADi, ROS 2, PX4, ArduPilot • Development & DevOps Tools: Docker, Continuous Integration, Git/GitHub • Hardware & Experimentation: UAV assembly, repair, and deployment • Motion-Capture: OptiTrack and Vicon • Languages: Spanish (Native), English (Native)		
EDUCATION	Georgia Institute of Technology 2022-2027 Ph.D. in Robotics <i>Doctoral Minor in Optimization Methods</i> • GPA: 4.0/4.0 • Advisor: Dr. Samuel Coogan • Thesis Proposal Approved (ABD) • Goizueta Fellow University of Texas at Arlington 2018-2022 Honors Bachelor of Science in Electrical Engineering <i>Minor in Mathematics</i> <i>Minor in Physics</i> <i>Certificate in Unmanned Vehicle Systems</i> • GPA: 3.9/4.0 Summa Cum laude • Full National Merit Scholarship Awarded • Electrical Engineering Honors Scholar		
WORK EXPERIENCE	Graduate Researcher AI for AutonomyMay 2026 - Present Sandia National LaboratoriesAlbuquerque, NM • Conducted applied research integrating artificial intelligence and advanced control methods for autonomous systems • Developed and evaluated algorithms for the perception, decision-making, and control of autonomous platforms • Implemented and tested methods in simulation and on hardware within a multidisciplinary research team Teaching Assistant Vertically Integrated Project in RoboticsJanuary 2024 - Present Georgia Institute of TechnologyAtlanta, GA • Aided professor in his instruction of this research-based course • Provided promising undergraduates with guidance performing applied research in the field of robotics • Taught control theoretical concepts in an accessible manner for undergraduates Graduate Research Assistant FACTS LabAugust 2022 - Present Georgia Institute of TechnologyAtlanta, GA • Advised by Dr. Samuel Coogan • Implemented novel aggressive tracking control methods on quadrotor hardware • Developed novel nonlinear control methods for safe autonomy • Published novel research at top conferences and journals in the fields of robotics and control theory		

Research Assistant | Autonomous Systems Lab

University of Texas at Arlington

November 2021 - May 2022

Arlington, TX

- Advised by Dr. Frank Lewis
- Received personal mentorship from a leading researcher in the field of controls and reinforcement learning (ranked 12th in the USA and 23rd in the world during my time in his lab)
- Aided graduate students on reinforcement learning research applied to quadrotor control

Teaching Assistant | Graduate-Level Intelligent Systems Course

University of Texas at Arlington

January 2022 - May 2022

Arlington, TX

- Aided professor in his instruction of the course
- Hosted review sessions for students in preparation for exams
- Gained teaching experience in a graduate-level course while still an undergraduate

Research Assistant | Dynamical Networks and Control Lab

University of Texas at Arlington

October 2019 - May 2022

Arlington, TX

- Advised by Dr. Yan Wan
- Gained experience with Robot Operating System (ROS), OpenCV, and path planning
- Original research in learning-based minimum time and energy path-planning for multi-vehicle systems

**SELECTED
PUBLICATIONS**

M. Cao, A. Harapanahalli, **E. Morales-Cuadrado**, and S. Coogan, “Trajectory Tracking for Systems with Unknown Time-Varying Disturbances.” Submitted to Open Journal of Control Systems. (*under revision*)

E. Morales-Cuadrado, “Computationally Efficient and Safe Control for Aerial Robotic Systems under Threat and Disturbance.” Ph.D. Thesis Proposal, Georgia Institute of Technology, 2026. [pdf available](#). [slides available](#).

E. Morales-Cuadrado, L. Chung, S. Kousik and S. Coogan, “RTD-RAX: Fast, Safe Trajectory Planning for Systems under Unknown Disturbances.” Joint Submission to 2026 Modeling, Estimation and Control Conference (MECC) and ASME Letters in Dynamic Systems and Control (ALDSC). (*under revision*) [preprint available](#).

E. Morales-Cuadrado, L. Baird, Y. Wardi and S. Coogan, “Lightweight Tracking Control for Computationally Constrained Aerial Systems with the Newton-Raphson Method.” IEEE Transactions on Control Systems Technology, 2026. (*accepted; to be published*) [preprint available](#).

L. Baird, **E. Morales-Cuadrado**, and S. Coogan, “Runtime Assurance for Uncertain Systems from Interval Signal Temporal Logic.” Submitted to IFAC Nonlinear Analysis: Hybrid Systems (NAHS), 2025. (*under revision*) [preprint available](#).

E. Morales-Cuadrado, C. Llanes, Y. Wardi and S. Coogan, “Newton-Raphson Flow for Aggressive Quadrotor Tracking Control.” [2024 American Control Conference \(ACC\)](#)

**RESEARCH
INTERESTS**

- | | |
|---------------------------------------|--------------------------------------|
| • Safe Autonomy | • Advanced Nonlinear Control |
| • Hardware Deployment | • Learning-Based Control |
| • Unmanned Ground and Aerial Vehicles | • Trajectory Generation and Planning |

**OPEN-SOURCE
CONTRIBUTIONS**

- **PX4 Autopilot:** Contributed documentation improvements to the official PX4 Autopilot manual.
 - **Vicon4PX4:** Developed and open-sourced a ROS 2 C++ Vicon–PX4 interface enabling external vision fusion for indoor quadrotor flight experiments; adopted and showcased by Georgia Tech’s Indoor Flight Lab as part of its official codebase. Also created a parallel package for OptiTrack systems.
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SERVICE & LEADERSHIP

Co-Founder and President of Puerto Rican Student Association

August 2024 - Present

Georgia Institute of Technology

Atlanta, GA

- Addressed the need for an organization to help Puerto Rican students feel at home and stay connected to our culture
- Formed a three-member co-founder board and identified a faculty advisor
- Raised funds for the organization and recruited members from campus
- Hosted professional, social, and cultural events to meet the needs of the growing Puerto Rican student population at Georgia Tech

Undergraduate Research Mentor

August 2021 - May 2022

University of Texas at Arlington

Arlington, TX

- As the lead undergraduate student Dr. Frank Lewis' Autonomous Systems Lab, I mentored junior undergrads in controls research and guided them on a research project in ground vehicle control
- Guided students in Robot Operating System, Python, Linux, and Control Systems

Teaching Assistant | Graduate-Level Intelligent Systems Course

January 2022 - May 2022

University of Texas at Arlington

Arlington, TX

- Aided professor in his instruction of the course
- Hosted review sessions for students in preparation for exams
- Graded homeworks, exams, and projects
- Gained teaching experience in a graduate-level course while still an undergraduate

Eta Kappa Nu Public Relations Chair & Circuits I Tutor

August 2020 - May 2022

University of Texas at Arlington

Arlington, TX

- Executive board member of UT Arlington's IEEE-HKN chapter
- In charge of social media advertising for honor society's events and outreach initiatives for the electrical engineering student body
- Volunteered to provide free tutoring as a service to electrical engineering students in Circuits I

HONORS AND AWARDS

Top-3 Finalist at a Deep Learning Research Symposium

November 2023

Professor-sponsored award for novel research in deep learning and robotics

Goizueta Fellowship at Georgia Tech

August 2022

Highly selective fellowship for graduate students of Latin-American descent

Summa Cum Laude Honors at University of Texas at Arlington

May 2022

Exceptional academic distinction in top GPA tier for graduation honors

Honors College Degree Requirements at University of Texas at Arlington

December 2021

Completed research projects in disparate fields throughout my undergraduate career with various professors and defended a senior capstone research project before the Honors College committee

Electrical Engineering Honors Scholar at University of Texas at Arlington

December 2021

Inaugural member of the electrical engineering honors cohort

IEEE-HKN International Electrical Engineering Honor Society Membership	August 2021
Selective undergraduate society recognizing outstanding academic achievement and community service	
Research Experience for Undergraduates Fellowship Recipient	August 2020
Fellowship to sponsor promising young undergraduate researchers	
Chance Vought Engineering and Science Endowment Scholarship	August 2020
Highly selective award given annually to three recipients across the entire College of Engineering	
Toray Composite Materials America Inc. Sons and Daughters Scholarship	May 2019
Academic scholarship awarded children of employees of Toray America	
James E. Casey Scholarship Recipient	January 2019
Academic scholarship awarded children of employees of UPS	
James R. Hoffa Memorial Scholarship Recipient	May 2018
Academic scholarship awarded children of Teamsters Union Members	
Grapevine, TX Rotary Club Scholarship Recipient	May 2018
Local rotary club scholarship for academic and civically-minded scholars	
National Hispanic Scholar	March 2018
Recognition of top national performers of Latin descent on the SAT	
National Merit Scholar	March 2018
Recognition of top national performers on the SAT	